

Benefits of forests in Cameroon. Global structure, issues involving access and decision-making hiccoughs

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Abstract

Among many other stakes, the economic stake derived from the exploitation of tropical forest resources is a burning issue. This is evidenced by insecurity in intergenerational access to forest resources and financial benefits relating to the latter, on the one hand, and by a deep iniquity at the intra-generational level, on the other hand. The following paper highlights, as a moral, social and policy dilemma, how stakeholders and generations, ‘self-interested’, mark out access to forest resources and to financial benefits relating to the latter. Through intensive participatory research, quantitative data collection, participant observation, future scenarios and some International Forestry Research’s social science methods and interactive games (SSM & IG) based on the evaluation of the sustainability of forest management systems, field research conducted in the forest zone of Cameroon on access to forest resources has generated two central results. Firstly, future generations will be confronted—in a dramatic way—to quantitative and qualitative scarcity of forest resources, following their over exploitation by present generations. Secondly, as concerns the intra-generational access benefits generated by commercial exploitation of forests and the assessment of the circulation of forestry fees, there is much inequity, in as much as those benefits are more profitable to a ‘forestry elite’—‘a self-interested block’—than to local communities, who strongly claim to have historical rights over these forests. As a contribution of social science to public knowledge and to policy development, this article is nourishing ‘rational choice’ and ‘rational egoism’ theory and is targeting decision-making processes in the ever first attempt of forest management decentralization and ‘legal’ benefits sharing in Central Africa (the second largest forest of the World). The article recommends the shortening of the distance between decision-making and beneficiaries, downwardly accountability, ‘bottom-up’ mechanisms of public dialogue in forest management and a collaborative infrastructure in the circulation and the distribution of forest benefits.

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1. Introduction

Commercial logging was inaugurated by the German colonization in Cameroon in the nineteenth

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century, as revealed by Schanz (1914). Under the French colonial period, it had registered a notable expansion motivated by metropolitan logics of financial accumulation (Meniaud, 1948). In 1920, for example, 59 000 tons of timber were exported from Cameroon to France. Moreover, according to Biesbrouck (2001), by 1948—already—the colonial administration granted French logging companies 25-year forest concessions covering more than 714 000 ha in the coastal region of the country. After a century of exclusive and authoritative regulation of forest management by the central state (Buttoud, 1991; Rice and Counsell, 1993; Oyono, 1995; Karsenty, 1999; Vabi et al., 2001), the change in forest policies outlined in the mid-1990s laid the legal foundations of the transfer of powers and responsibilities to peripheral actors. The forestry Legislation enacted in 1994 and the subsequent provisions—the most innovative experiment in the Central Africa region (Nguinguiri, 1999)—are supposed to set up the institutional basis of a decentralization process. Today, it is more and more highlighted in the domestic scientific production that one of the major achievements of the decentralization of forest management in Cameroon is related to access to financial benefits. The notion of benefits, in the logic of this paper, will include the resource itself and financial benefits or compensations allocated for its exploitation, thereafter called fees.

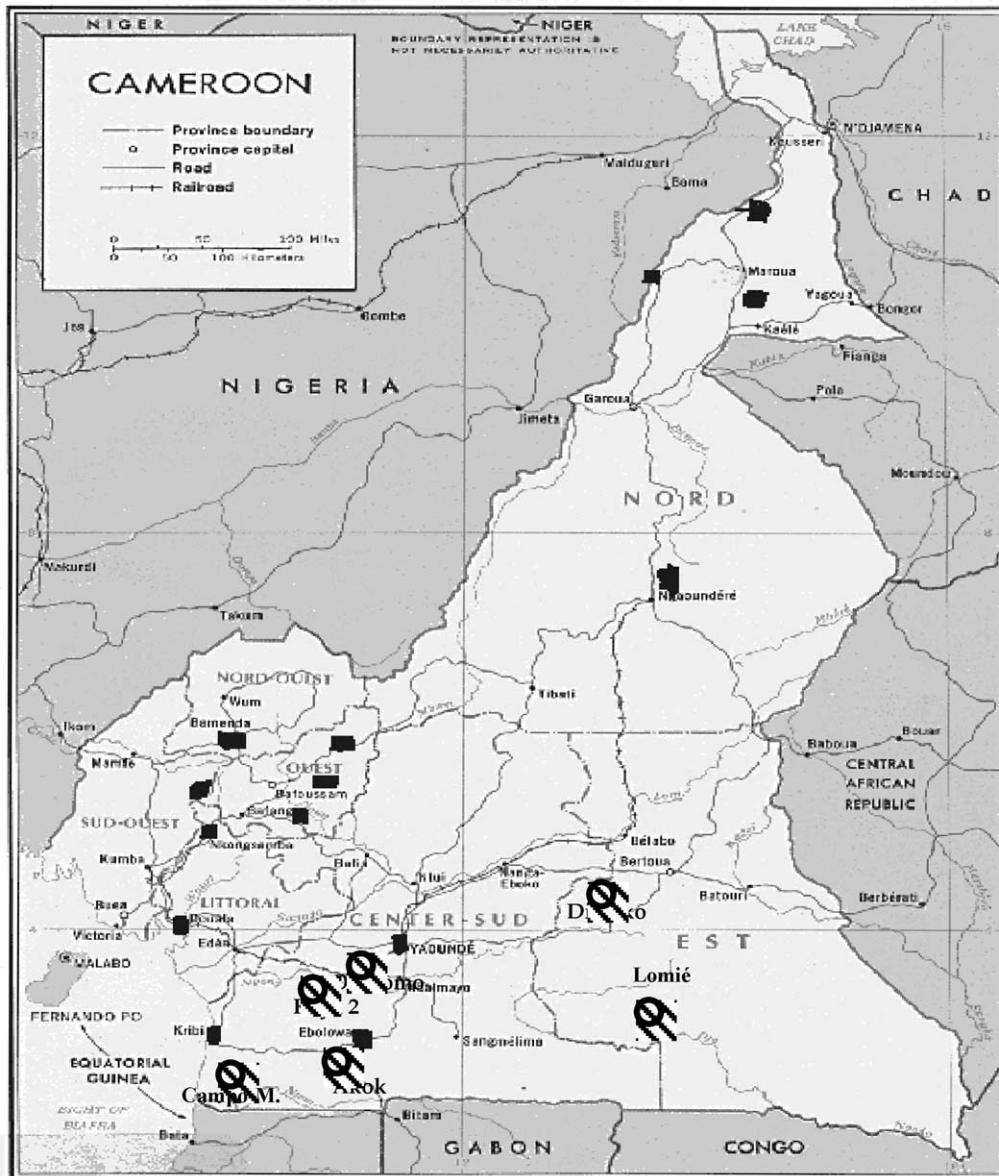
This paper is based on empirical results of research work carried out between 1998 and 2000 at the Center for International Forestry Research (CIFOR, Central and West Africa). The methods used combined (i) interactive ‘games’ and tools conceived by CIFOR, to assess human well-being as an outcome of sustainable forest management (Colfer et al., 1998), including the assessment of stakeholders’ benefits, in-depth stakeholder analysis, historical transects of the landscape, the assessment of stakeholders’ rights to manage the forest; (ii) in-depth historical analysis; (iii) an analysis of local communities’ discourse and representations; (iv) future scenario of access to resource; (v) classical quantitative surveys. The article reconstitutes the structure of access to resource and the outline of social, political and strategic distribution of forestry fees introduced by the 1994 Legislation. It weighs

the ‘trans-generational’ dimension of access to forest benefits, and the policy choice consisting in allocating forestry fees to local communities throughout local governments (or municipalities). It tries to reproduce the guidelines of insecurity and imbalance governing access to forest benefits—in the large sense—and assesses the distribution of forestry fees in Southern Cameroon. This analysis will finally lead to political, administrative and scientific lessons that may serve as instruments for decision-makers, law-makers, planners and for policy research (Scheme 1).

2. Background

The issue of having access to the benefits generated by the exploitation of natural resources has been increasingly attracting attention (Fauchaux and O’Connor, 1999; Peluso, 1992; Zerner, 2000; Watts, 2000), on the one hand. On the other hand, significant emphasis is placed on obligations of the present generations towards the future generations (Colfer et al., 1998; Dobson, 1998). Even in the ‘African wisdom’, it is suggested to social agents ‘to eat today and think of tomorrow’. This issue is also captured in its intra-generational aspects (Bonnieux and Ranielli, 1999; Schroeder, 2000; Vandergest, 2003), the whole school of ‘new institutionalism’—when these resources/benefits are under ‘common property regime’ (Ostrom, 1990; Wade, 1987)—and streams of political ecology (Sachs, 1993; Lowe, 2000; Martin, 2002). At the domestic level, it is documented that access to forest benefits is a burning issue (Bigombé, 2003; Oyono, 2003).

Research conducted in Cameroon since 1998 reveals both inequity and insecurity in the gradual access of human generations to forest resources and benefits (Tchikangwa et al., 1998; Diaw et al., 1998). In addition, it has been noted that there is also, at a given time, a remarkable imbalance in access of various stakeholders to forest financial benefits (Tchikangwa et al., 1998; Oyono and Diaw, 1998; Kouna, 2001; Fomété, 2001; Bigombé, 2003). Some other works point out the permanence of conflicts around those stakes (Verhagen and Eindhoven, 1993; Bigombé and Dabiré, 1999; Oyono,



Scheme 1. The research location in the rainforest zone of Cameroon

2002). Egbe (1997), Diaw (1997), Vabi et al. (2001) and Oyono (2001) have identified legal and socio-logical constraints to self-governance of forest resources and benefits. Thus, given its importance, the issue of access to forest benefits is continuously tackled all over the world: this article is a contribution to that effort.

3. The structure of inter-generational access to forest benefits

The application of CIFOR's social science methods on forest sustainability and on human well-being assessment, in particular exercises focused on the evaluation of 'security in the intergenerational access

to forestry resources', games presenting 'the sharing of benefits accruing from forest exploitation', historical transects and participatory mapping have led to a general view of the issue. For the assessment of security in intergenerational access to forest resources, 20 respondents selected among stakeholders in presence were, each, asked to distribute, to successive generations, 100 pebbles representing forest resources. In general, 'local communities', the 'state', 'logging companies', 'medium-scale farmers' and 'agricultural research and NGOs' have been identified as key stakeholders. This survey was conducted in 10 research sites: results presented here are from four zones assembling characteristics of the whole research area, Akok in the South province, Kaya and Koli/Nkongabok in the Center province and Dimako in the East-province (Fig. 1).

Another objective of this exercise is to set up a 'future scenario' (Wollenberg et al., 2000). Future scenarios are 'stories' or a materialization of the mental representation of 'what might be', especially

Stakeholders' perception of intergenerational access to forest resources

In the Akok area (●) past generations have exploited 36% of resources, while 45% are exploited by present generations and future generations will find the remaining 19%. The figure reveals that in Kaya2 (■) 22% of forest resources have been taken by past generations, the living (present generations) are exploiting 43% and future generations will probably exploit 35%. In Koli-Ngoumou (▲), present generations are taking the 'share of the lion' (56%) of resources, past generations are supposed to having exploited 30%, while future generations will find only 14%. These distributions give way to a collective answer—a consensus in the above case—around the social and the historical distribution of resources between '*those who lived yesterday, those who live today and those who will live tomorrow*'. The conclusion is that there are few intergenerational equity in access to—and exploitation of—forest resources.

in forest resources management, where uncertainty and complexity are high. Discussions organized with various stakeholders in the study areas, with local communities at the first place, bring systematically up the fact that 'those of today' (present generations) are exploiting forest resources viewed as a bio-physical stock more than 'those of tomorrow' (the future generations) and more than ever 'those of yesterday' (the past generations). The result of the exercise based on 'intergenerational' access to forest resources have been each time confirmed by the implementation of other tools, as historical transects of landscape drawing the forest as it could be under 'those of yesterday', as it is today (under the living) and as it might be under 'those of tomorrow'.

This said, the results of all these exercises put forward inequity and insecurity in intergenerational access to forest resources, which are in the long run used for good to the detriment of future generations. These results have some explanatory variables. It is a persistent trend. Social inequity and unfairness in access to resources can be explained in general by human pressures on forest, for reasons of generating financial incomes through agriculture and exploitation of non-timber forest products, within a context of deep economic recession, as documented in Cameroon by Tchoungui et al. (1995), Oyono (1998), Ndoye et al. (1998) and Mala (2001). Rapid technology change—for example, sophisticated production tools in timber harvesting—is also a major driving force of this 'future insecurity'. Olé (1998) estimates that an area that could be logged during 1 year in the seventies can be logged nowadays within 3 months. Other variables include the lack of law enforcement, the gaps in legal provisions related to excessive timber harvesting, the lack of agricultural supply in rural areas, growing population and easy access to both national and international market (Buttoud, 1991, for timber trade).

4. Intra-generational access to forest benefits in Southern Cameroon

4.1. The state and logging companies as main beneficiaries

The visualization of actors involved in the question of access to forest benefits reveals, from one geo-

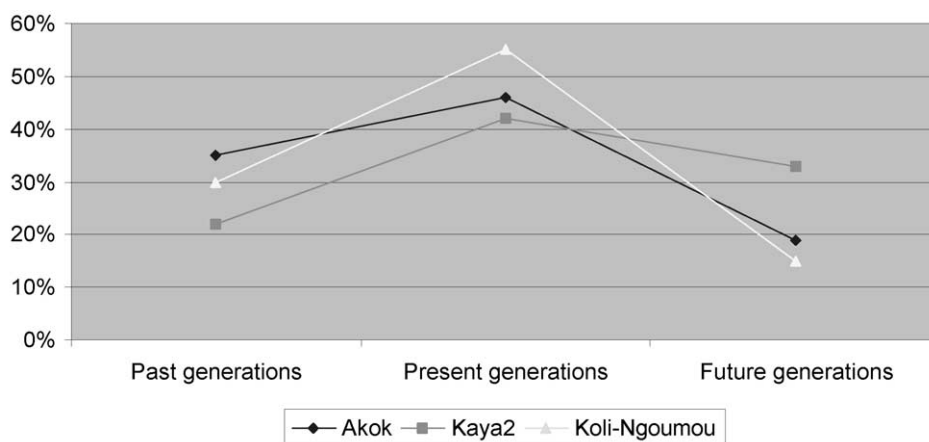


Fig. 1. Distribution of 100 pebbles representing forest resources among three 'blocks' of generations in three research areas.

graphical area to another, a number of stakeholders connected by a set of strategic, social, financial and political stakes (Verhagen and Einthoven, 1993; Bigombé and Dabiré, 1999; Kouna, 2001; Fomété, 2001; Oyono, 2002). These stakeholders are divided into 'functional categories': (a) the 'rightful claimants' or the 'owners' of the forest; (b) the 'users' of the forest and (c) the 'decision-makers' about the forest. Another CIFOR's social science method assessing human well-being in forest management gives a global view of intra-generational access to forest benefits. Its scientific objective is to quantify, through a distribution exercise, stakeholders' perception of this access. Firstly, it is asked of respondents to identify, on the basis of the surrounding forest conditions, the major forest benefits. In general, 'money', 'timber', 'game animals', 'crops', 'medicinal products' and 'other non-timber products' came at the top of the list in all the research sites as categories of

forest benefits. Secondly, as in the perception of intergenerational access, respondents are asked to distribute with the aid of 100 pebbles, which is supposed to be the share of each identified stakeholder in each identified category of benefit. The scores are then added up and divided, for each category of benefit, by the number of respondents. Quantitative results of the exercise are presented by benefits and by stakeholders (Table 1).

Distribution lay emphasis on the leading position of the state and logging companies on benefits like 'money' and 'timber'. The fact that timber companies and the state have the 'share of the lion' for these two benefits is significant. It is in accordance with the colonial and postcolonial logics giving 'accumulation rights' to the state and its partners (Meniaud, 1948; Karsenty, 1999) and 'use rights' for any other purpose to local communities. Local communities perceptions of 'who takes what from the forest' within the present

Table 1

Intra-generational distribution of main forest benefits between stakeholders at Koli-Ngoumou area

Stakeholders benefits (%)	The state	Municipality	Village community	Logging companies	Neighboring villages
Money	25	15	15	40	5
Timber	00	00	30	60	10
Game animals	00	00	50	20	30
Crops	00	00	70	20	10
Medicinal products	10	00	45	25	20
Other 'NTFP' ^a	10	00	40	35	25

^a Non-timber forest products.

generations does not result only from a historical crystallization or frustration (Diaw and Njomkap, 1998; Karsenty, 1999) and ‘ecoactivists’ (Kuwik, 1996; Zinga, 1996; Mimbimi, 1996). It is a reproduction of the reality. And that issue is not thus settled at random. Without going too far, they have visible proofs every day (vehicles, production tools, trucks and other material symbols). The availability of information, through the democratization of the public life since 1990, on invested capital and part of revenues collected by logging companies, indigenous operators ‘living’ along the networks of timber exploitation and by the state facilitate the understanding of the economic status and the weigh of each of the two stakeholders. This is an obvious and one of the less significant example: Ndjatsana and Ndjodo Nga (1998), in a case study on a second level timber company, have revealed that only the annual wage bill (for foreigners and nationals) was 200 000 USD. At the same time, 60 000 USD of taxes were annually paid by that small company to the state. In 1998 only, 3 000 000 m³ of timber have been extracted by timber companies from the Cameroonian rainforest, in a very competitive and frenetic way (Olé, 1998). Distributions done by respondents and data like these ones above show that access to forest benefits within the present generations or the ‘living’ is based on iniquity and on transversal insecurity. In the collective imagery of local communities, forest—strongly considered a historical and unalienable good—is more profitable to ‘others’ than to themselves.

The creation of protected areas all along the 20th century is also an outstanding illustration of intra-generational inequitable access to forest benefits, mainly resources as bio-physical units. According to the new forestry Legislation of 1994, protected areas—national parks, game reserves, forest reserves, botanical gardens, etc.—should cover 30% of the national territory. One research site, Koli/Nkongabok, in the Ottotomo Forest Reserve, highlights the disparity induced by the creation of protected areas. When this Reserve was demarcated, local communities were not involved or even consulted. In a given segment of time, and within ‘those of today’, access to its resources has always been prohibited for natives. In the meantime, the colonial state and the postcolonial state have three times opened the Reserve to ‘legal’ timber exploitation, in 1946, 1976 and 1993

(ONADEF, 1998; Jum et al., 2001), to worsening inequity on the occasion. Accordingly, the co-existence of the state and local communities on protected areas lies on a ‘legal dualism’ and a ‘conflict of language’. On the one hand, local communities continue to claim their ownership over the pieces of forest adjacent to their villages; on the other hand, the state representatives brandish law in force. The zoning plan of Southern Cameroon, attributing 80% of lands to the state is nourishing that conflict of language since its design in 1993.

4.2. Illustrations on intra-generational access to forest royalties

Socio-economic development and social justice are among the reasons officially evoked for the implementation of a forest management decentralization model in Cameroon. To that effect, reforms in the forest sector have generated a ‘fiscal decentralization’ (Smoke, 2000; Carret, 2000), among other basic instruments. The management of these instruments is laid down by official texts that clearly indicate their destination, their circulation and their distribution norms. According to the forest Legislation in force, timber companies must, in a scale, pay annual royalties while exploiting a timber concession. The state takes 50% of the royalties; 40% goes to the rural council (or *commune*¹) concerned and 10% to surrounding villages. In principle, the 40% of the global amount of forest royalties allocated to the commune should be channeled to micro-projects at the grass-roots level. In addition, an amount of 1.5 USD by cubic meter of timber, a sort of ‘ecotaxes’, should be allocated to each village accommodating timber exploitation in its ‘forest’ (Milol and Pierre, 2001). This category of compensation goes directly to village communities. The following paragraphs reproduce some of our research evidences on access to forest royalties in the Dimako region, East province of Cameroon.

Group discussions with local communities during this research have ultimately been transformed into a

¹ In former French colonies, the commune or the ‘municipality’ is a territorial unit managed by an elected body, the mayor and the municipal council. It is the only existing type of local government in Cameroon.

‘political and moral trial’. For local communities, the intra-generational share of annual royalties liberated by timber exploitation between the state, the commune and themselves is simply unfair. They argue that as ‘owners’ of the forest—in their logics—they merit more than 10%. In short, they believe and demonstrate that they are in a situational prism of ‘rich forest, poor people’ (Peluso, 1992). In Cameroon, allocation of forest financial benefits has adopted a metamorphosing configuration since 1994. At the beginning, they were directly given to the local communities, which could manage them as they pleased. Sometimes, they invested the money in the promotion of community well-being² and, often, they took a ‘revenge on history’, by *‘eating and drinking with the money of their forests’*. This form of local ‘governance’ and related ‘political and social power formation’ (Habermas, 1996) have been deeply criticized by the ‘dominant system’ (officials of the ministry of Forests, administrative authorities and municipal authorities), putting forward the argument of ‘unconscious’ use of timber compensations. The government reacted in 1998, through an administrative provision, a ministerial order, by re-centralizing the destination of the local communities’ portion of revenues—and the planning of their management—at the commune level.³

This administrative provision says that forest royalties management committees must be created at the level of each ‘sub-division’ or ‘district’ accommodating timber exploitation activities. In the Cameroonian administrative organization, a sub-division refers to the regional level. This regional committee is placed under the supervision of an administrative authority (the *sous-prefet*) and is presided over by the mayor or his representative. In addition, at the level of each village (or group of villages) adjacent to a forest concession, village management committees, headed by a chairman and four or five members, have been created for forest royalties, as indicated by the very ministerial order. With the re-centralization of the management of forest royalties, from the village level to the commune level, the regional committee has become, de jure, the manager of the said royalties.

Nowadays, it is asked to village management committees to identify local level actions to be funded, such as schools, health centers, etc. But it is the regional committee that has the power to liberate funds for any village level micro-project. As results of surveys carried out by Kouna (2001), communes or municipal authorities very often create, on paper, fictitious projects to be funded and thus misappropriate the money. From the peasant point of view, the state does not want to share with them: that is to say, it *‘takes back with the left hand what it gives with the right hand’*. Furthermore, a ‘strategic axis’ (Oyono, 2001) made up by members of village management committees, municipal authorities, administrative officials and timber companies has emerged in that process. According to local communities in Southern Cameroon, the unconfused goal of this axis is to keep forest royalties at the regional level in order to share them or to mismanage them. In the second category of forest royalties, the ‘1.5 USD by cubic meter’, indeed money is allocated to villagers themselves, but the way the principle is operationalized gives rise to some questions: (i) Given the fact that villagers do not have the required skills, are they involved in the counting of cubic meters? (ii) If this is done by timber companies workers, do they report the exact amount to villagers? Mendouga (1998) has concluded that the principle of 1.5 USD is not applied as it should be and that generally timber companies give any amount they want to local communities.⁴

The creation and the inflation of village management committees in the whole Southern Cameroon has built a new social differentiation and stratification, in the sense that members of these organizations are being integrated in the circle of village authorities. In the principles governing access to common resources in these societies, externally created organizations or groups have no ‘constitutional’ legitimacy, and therefore no rights (Diaw and Oyono, 1998; Oyono, 1999; Djeumo, 2001). The new ‘forestry elite’, in other words members of these village committees, is formed mainly by retired civil servants and young men excluded from towns by the exhaustion of the ‘urban

² Kouna (2001) talks of health centers and classrooms in most cases.

³ See joint Order No. 00122/MINEFI/MINAT of 29/04/1998.

⁴ Strategically, timber companies jump on that opportunity. The fact that after so many ‘frustrations’ local communities want first *‘to eat and drink’* with money provided by forest exploitation is in their advantage.

model' in a context of economic recession. In the light of this, the ministerial order asking villagers to set up management committees has opened up to various forms of legal 'coups d'état' or to an officially authorized usurping of powers, by these 'self-interested' groups (Rutherford, 1996).

As pointed out above, many committees have operated a shift towards regional political elites, administrative authorities and logging companies managers, to be part of the 'strategic axis' focused on forest revenues. And more and more, they really put in practice the notion of the centralization of authority in group decision-making. Oyono (2001) and Bigombé (2003) show that, in a free-riding process, committees are disconnected to the rest of the people and do not account to them about issues such as royalties manipulation at the regional level and their interactions with external actors. According to Karsenty (1999), the functional aim of this axis is to marginalized local communities from having a meaningful voice on the destination of forest royalties. It is reported by Efoua (2001) and Assembe (2001) that committee members are regularly accused of taking important amounts of money for their silence. For these two authors, *communes*, *sous-prefets* and logging companies always make the tactical choice consisting in giving something to committee members instead of transferring the whole amount of money to local communities.⁵ Such a context generates marked intra-generational imbalances.

As a matter of fact, such a situational structure contributes to extremely extending the potential of intergenerational conflicts on the issue of benefits accruing from the exploitation of forests in Cameroon. Therefore, in many social localities, conflicts have emerged this last decade between various stakeholders and lasted for a long time. The most significant has been opposing local communities to logging companies. In many parts of Southern Cameroon, local communities regularly and openly target logging companies in their protests of the 'forestry order'. In these circumstances, logging trucks are barricaded, while workers and their patrons are kept locked in

forestry camps. Sometimes, these situations require the intervention of a military troop (Mimbimi, 1996). Whether spontaneous or planned, such micro-located eruptions unify the implicit and explicit stakes involved in the management of forests. These eruptions are, as a matter of fact, significant and show that ongoing decentralization is more an 'authoritarian modernism' (Scott, 1998) in forest management and not a project of local communities' empowerment.

5. Impact on local development and ecological sustainability

If the decentralization process has temporarily reduced the charge of impulses oriented by village communities towards the state, it has on the contrary transferred the potential of conflicts to the village arena. The assessment of the new vertical configuration of forest royalties management—from the communes to villages—concludes to the scarcity of micro-level community projects. To that effect, Kouna (2001) and Bigombé (2003) report that in the East and the South provinces of Cameroon, socio-economic outcomes of the decentralized management of forest royalties are extremely weak. Only 10% of villages concerned by that issue—in 22 *communes*—enjoy some welfare benefits. In the Dimako region, one of those regions where the issue is a burning one, official documents consulted show that, for the years 1998/1999 and 1999/2000, socio-economic achievements represent only 40% of the 53 000 USD intended for village communities.

The allocation of forest royalties to local communities is qualified by the official rhetoric as an appropriate weapon in the fight against poverty. At the village level, there are no meaningful indicators on the achievement of such an objective. On the one hand, when annual forest royalties are given directly to villagers (before the ministerial order of 1998), they decide to 'eat and drink now'; on the other hand, the transfer of these royalties to the *commune* level leads to mismanagement and embezzlement options involving *commune* authorities, *sous-prefets* and village committee members. That is why local communities, with young people at the front line, believe that they must do whatever they want with their small portion of money, and thereon make the collective choice of

⁵ Assembe (2001) reports that in one of the villages of the Dimako research site, the president of a village committee was accused of funds embezzlement: the *sous-prefet* asked the village community not to take him to justice.

eating and drinking now, as a continuous message sent to the state and a response to the dominant system.

Institutional economics explanations would add ‘rational behavior’ of social agents and stakeholders as self-interested individuals and groups not feeling any obligation towards future generations. In that view, ‘those of today’ act for their own interest and, obviously, do not care about ‘those of tomorrow’. To justify such a ‘rational choice’, young people in many villages of Southern Cameroon react recurrently as follows: *‘Others—the past generations, the state and timber companies—have already eaten too much from these forests, it is now our turn’* (Efoua, 2001). In sum, and beyond all the explanatory variables/approaches, the domestic potential of human influence on forests in Cameroon has been considerably expanded in the last five decades as human societies moved from a semi-contemplation attitude towards resources to the implementation of forest-based operations more and more motivated by financial accumulation.

There are prominent trends—the choice consisting in just eating and drinking with money of forest, for example—highlighting the fact that villagers have opted to an eschatological vision of the issue forest management in the whole. There is no future for forest royalties. Moreover, in the light of aggregate facts revealing iniquity and iniquity in both intergenerational and intra-generational access to benefits, forest as a bio-physical entity should not equally have a future, in the local perspective. From an individual and collective point of view, forest resources and benefits are considered by local communities as instruments of social and economic mobility, for the ‘others’ and not for themselves. This is why they are favorable to the attribution of more forest concessions to timber companies, if they could, after so much time, take a little advantage.

‘Legal dualism’ in the ownership of forests⁶ transformed into protected areas since the colonial period and the insecurity it has generated among stakeholders within past and present generations equally create, from end to end, ‘desperate strategies’—like serious encroachments recorded in many forest reserves—of ‘no forest at all for nobody’ built in the last two decades by local communities in Southern Cameroon. To that effect, surveys carried out by Tiani (2001) at Koli/Nkongabok in the Ottotomo Forest Reserve

reveal that despite prohibition measures, the dense forest have been reduced by 45% from 1960 to 2000, while field crops area have increased by 60%. All these fragmented eschatological practices are meaningful signs of ecological uncertainties and augurs well for the spectre of ‘forestry nothingness’ brandished by local communities under intra-generational iniquity conditions. In short, more radical individuals among local communities, young people for example, wish the ‘end of forests’, because in that situation, the state and timber companies will be the great losers.

Some of the methods used, like the SSM & IG of CIFOR and future scenarios, are innovative. They have been helpful for the qualitative aspects of the research and for the capture of stakeholders’ perceptions, representations and discourse. Historical transacts of the landscape that appeared as an exciting exercise though results obtained therefrom should be handled with reservations. Too many discords have emerged amongst stakeholders’ perception of many items of investigation. The quantitative component of fieldwork finally appeared as marginal. In sum, while some of the SSM & IG need to be improved and sharpened, the methodology used, located at the intersection of ‘what people think’, on the one hand, and ‘what is’, on the other hand, have been relevant for the capture of the issues tackled by the study.

6. Conclusions

As it can be noted from the previous pages, the imbalances identified in access to forest resource and forest financial benefits are embedded in policies, laws, regulations and orders enacted by the state since the colonial period. If the issue has policy and decision-making roots, attempts of solutions should accordingly be found in the policy and decision-making spheres. Forest royalties should be directly and fairly given to local communities themselves. In that sense, efforts are needed in the design of strong mechanisms aiming at (i) putting an end to impunity surrounding the ‘forestry channel’ that goes from ministries (Forests, Finance and Interior) to local communities, passing by logging companies, *communes*, *sous-prefets* and village management committees; (ii) enacting laws and regulations based on the obligation of downward accountability at all the levels

of the ‘forestry channel’; and (iii) promoting and enhancing good decision management.

The ‘re-decentralization’ of the management of forest financial benefits may seem as a huge task. Our research has sufficiently observed that the failure of this local management has increasing extra-local roots, like policy, decision-making and implementation hurdles. It equally lies at the confluence of local level explanations: these include low organizational instruments, low institutional arrangements for the community management of resources, ‘free riding’ behaviors, lack of strong social agreements. Nevertheless, these weaknesses can be tackled by NGOs’ expertise, as a supply side of local capacity building. People participation and local community empowerment should not remain a simple rhetoric or a cliché, as pointed out by [Buchy and Hoverman \(2000\)](#). Any change of policies of this type should be based on methods of experimentation and on a conscious approach. A small portion of annual taxes paid to the state by logging companies can be liberated for contracts with NGOs. Many logging companies’ managers believe that it is fair to allocate directly the royalties to local communities, not because they believe too much in equity but because they want to avoid regional authorities such as municipal authorities, *sous-prefets* and Forest department officers, considered as highly corrupted.

Iniquity in intergenerational access to forest benefits is not disconnected to the structure of intra-generational access to the same benefits. If relevant policy and decision-making changes can impact on intra-generational access to these benefits, as stressed above, security in intergenerational access depends on long-term policy implementation. Empirical results of this research are showing that present generations are benefiting more than past generations and future generations. Within present generations, it appears that timber companies and the state have the ‘share of the lion’ on money and on timber. This tendency can be transformed positively by a political willingness aiming at innovating and disseminating appropriate policies and strategies for long-term management of forests. In such a process, policy makers should explore new models, like collaborative management for example, bringing local communities into the arena, giving them a voice in decisions concerning the present and the future of forests and inherent

benefits. Of course, policy change and law enforcement is one thing and their application is another thing.

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References

- Assemble, S., 2001. La démocratie locale dans les organisations nées de la décentralisation de la gestion des ressources forestières au Cameroun. CIFOR, Yaoundé.
- Biesbrouck, K., 2001. New perspectives on forest dynamics and the myth of ‘Communities’: reconsidering co-management of rain-forests in Cameroon. IDS Bulletin 33 (1), 55–64.
- Bigombé, P., 2003. The decentralized forestry taxation system in Cameroon. Local Management and State’s Logic. WRI, Washington, DC.
- Bigombé, P., Dabiré, B., 1999. La gestion des conflits liés à la gestion des ressources forestières au Cameroun: Etat des lieux (1994–1999). FTTP-Cameroon, Yaoundé.
- Bonnieux, F., Ranielli, P., 1999. Sustainable development: a contribution of economic theory. First International Symposium on Sustainable Ecosystem Management, Chambéry.
- Buchy, M., Hoverman, S., 2000. Understanding participation in forest planning. A review. Forest Policy and Economics 1, 15–25.
- Buttoud, G., 1991. Les Bois africains à l’épreuve des marchés tropicaux. ENGREF, Nancy.
- Carret, J.-C., 2000. La réforme de la fiscalité forestière au Cameroun. Débat politique et analyse économique. Bois et Forêts des Tropiques 264 (2), 37–53.
- Colfer, J.C.P., Wadley, R.L., Harwell, E., Prabhu, R., 1998. Intergenerational access to resources: developing criteria and indicators. CIFOR, Bogor.
- Diaw, M.C., 1997. Anthropological institutions and forest management. What institutional framework for community-based management in Cameroon? WWF-CPO, Cameroon.
- Diaw, M.C., Njomkap, J.C., 1998. La terre et le droit: Une anthropologie institutionnelle de la tenure coutumière, de la jurisprudence et du droit forestier chez les peuples Bantou et Pygmées du Cameroun méridional forestier. INADES, Yaoundé.
- Diaw, M.C., Oyono, P.R., 1998. Dynamiques et représentations des espaces forestiers au Sud-Cameroun. Pour une relecture sociale des paysages. FTTP Bulletin 15/16, 26–33.
- Diaw, M.C., Oyono, P.R., Sangkwa, F., Bidja, C., Efoua, S., Nguiébouri, J., 1998. Social science methods for assessing criteria and indicators of sustainable forest management: a report of the tests

- conducted in Cameroon humid forest benchmark and in the Lobé and Ntem rivers basins-Part 1. IITA/CIFOR, Yaoundé.
- Djeumo, A., 2001. Développement des forêts communautaires au Cameroun: genèse, situation actuelle et perspectives. DFID/ODI, Londres.
- Dobson, A., 1998. Justice and the Environment: Conception of Environmental Sustainability and Theories of Distributive Justice. Oxford University Press, Oxford.
- Efoua, S., 2001. Etude des comités de gestion des redevances forestières dans l'arrondissement d'Ebolowa. CIFOR, Yaoundé.
- Egbe, S., 1997. Legal constraints to community-based management in Cameroon. WWF-CPO, Yaoundé.
- Faucheux, S., O'Connor, M., 1999. Natural capital: a controversial concept. First International Symposium on Sustainable Ecosystem Management, Chambery.
- Fomété, T., 2001. La fiscalité forestière et l'implication des communautés locales à la gestion forestière au Cameroun. DFID/ODI, Londres.
- Habermas, J., 1996. Between Facts and Norms. Contributions to a Discourse Theory of Law and Democracy. MIT Press, New Baskerville.
- Jum, C., Oyono, P.R., Tiani, A.M., 2001. Assessing the context of 'outside' and 'inside' management of the Ottotomo forest reserve. CIFOR, Yaoundé.
- Karsenty, A., 1999. Vers la Fin de l'Etat forestier? Appropriation des Espaces et Partage de La Rente Forestière au Cameroun. Politique Africaine 75, 14–23.
- Kouna, C., 2001. Décentralisation de la gestion forestière et développement local: performance et accountability dans la gestion locale des revenus forestiers à l'Est-Cameroun. CIFOR, Yaoundé.
- Kuwik, B., 1996. Cameroon: le combat inégal. Courrier de La Planète 35, 21–26.
- Lowe, C., 2000. Global markets, local injustice in Southeast Asian seas: the live fish trade and local fishers in Togean Islands of Sulawesi. In: Zerner, C. (Ed.), People, Plants, and Justice. The Politics of Nature Conservation. Columbia University Press, New York, pp. 234–254.
- Mala, A.W., 2001. The context of Forest Agriculture Interface and the Dynamics of Forest Agriculture Changes. CIFOR, Yaoundé.
- Martin, G., 2002. Grounding social ecology: landscape, settlement, and right of way. Capitalism, Nature, Socialism 13 (1), 3–30.
- Milol, A., Pierre, J.-M., 2001. Impact de la fiscalité décentralisée sur le développement local et les pratiques d'utilisation des ressources forestières au Cameroun. Banque Mondiale, Yaoundé.
- Mimbimi, E., 1996. Conflit entre les populations de Meyo-Centre, les Compagnies forestières UTC/Polywood et l'Etat. SAILD/APE, Yaoundé.
- Mendouga, L., 1998. Comment les populations forestières gèrent-elles les 1000 frs par mètre cube? FTTP Bulletin 15/16, 16–22.
- Meniaud, J., 1948. Le développement de l'exploitation forestière au Cameroun. France Outremer 227, 160–167.
- Ndjatsana, M., Ndjodo Nga, T., 1998. L'exploitation de l'Azobé au Cameroun: Etude de cas de la Société 'Design'. CIPCRE/ICCO, Yaoundé.
- Ndoye, O., Ruiz Pérez, M., Eyebé, A., 1998. The Market of Non-Timber Forest Products in the Humid Forest Zone of Cameroon. ODI, London.
- Nguingui, J.C., 1999. Les approches contractuelles dans la gestion des écosystèmes forestiers d'Afrique centrale. Revue des initiatives existantes. CIFOR/CORAF, Pointe-Noire.
- Olé, Z., 1998. Mécanismes de collaboration cohérente entre les parties concernées par l'exploitation forestière au Cameroun. CIPCRE/ICCO, Yaoundé.
- ONADEF, 1998. Plan d'aménagement de la Réserve Forestière d'Ottotomo. ONADEF, Yaoundé.
- Ostrom, E., 1990. Governing the Commons. The Evolution of Institutions for Collective Action. Cambridge University Press, Cambridge.
- Oyono, P.R., 1995. Les acteurs de l'exploitation forestière industrielle au Cameroun. SAILD/APE, Yaoundé.
- Oyono, P.R., 1998. Cameroon rainforest. Economic crisis, rural poverty, biodiversity. Ambio 27 (6), 557–559.
- Oyono, P.R., 1999. Institutional disorders and the social feasibility of community forests in Cameroon. An overview and some field lessons. IITA/CIFOR, Yaoundé.
- Oyono, P.R., 2001. Infrastructure organisationnelle et gestion décentralisée des forêts au Cameroun. Eléments d'anthropologie écologique et leçons intermédiaires. CIFOR/WRI, Yaoundé.
- Oyono, P.R., 2002. Characterizing conflicts of access to forest benefits. Legal 'gangsterism' and Panarchy in Southern Cameroon. International Association for the Study of Common Property Meeting, Harare.
- Oyono, P.R., 2003. Institutional paralysis, representation and decentralized forest management in Cameroon. Elements of natural resources sociology for social theory and public policy. WRI, Washington, DC.
- Oyono, P.R. et Diaw, M.C., 1998. Méthodes des Sciences Sociales sur la Durabilité des Forêts dans le Sud-Cameroun: Rapport des Tests conduits dans le Benchmark de la Forêt Humide et dans les Bassins de la Lobé et du Ntem- 2ème partie. CIFOR/IITA, Yaoundé.
- Peluso, N.L., 1992. Rich Forests, Poor People: Resource Control and Resistance in Java. University of California Press, Berkeley.
- Rice, T., Counsell, S., 1993. Forest Foregone. The European Community's Trade in Tropical Timbers and in the Destruction of Rainforests. Friends of the Earth, London.
- Rutherford, M., 1996. Institutions in Economics. The Old and the New Institutionalism. Cambridge University Press, Cambridge.
- Sachs, W., 1993. Global ecology and the shadow of 'Development'. In: Sachs, W. (Ed.), Global Ecology. A new Arena of Political Conflicts. Zed Books Ltd, London, pp. 4–21.
- Schanz, M., 1914. Le Régime Forestier dans Les Colonies Allemandes. Bibliothèque Coloniale Internationale, Hamburg.
- Schroeder, R.A., 2000. Beyond distributive justice: resource extraction and environmental justice in the tropics. In: Zerner, C. (Ed.), People, Plants, and Justice. The Politics of Nature Conservation. Columbia University Press, New York, pp. 52–66.
- Scott, J.C., 1998. Seeing Like a State. How Certain Schemes to Improve Human Condition Have Failed. Yale University Press, New Haven.
- Smoke, P., 2000. Fiscal decentralization in developing countries: a review of concepts and practices. United Nations Research Institute for Social development, Ottawa.

- Tchikangwa, N.B., Sikoua, S. Metomo, M. et Adjudo, M.F., 1998. Test des Méthodes des Sciences Sociales de Vérification des Critères et Indicateurs d'Aménagement Durable des Forêts: Périphérie-est de la Réserve du Dja (Sud-Cameroun), CIFOR Report, Bogor (Indonesia), Bogor.
- Tchoungui, R.S., Gartlan, S., Mopé Simo, J.A., Fondo, S., Ndjatsana, M., Winpenny, J., 1995. Structural Adjustment and Sustainable Development in Cameroon. ODI, London.
- Tiani, A.M., 2001. Etude de contexte dans la Réserve Forestière d'Ottotomo. CIFOR, Yaoundé.
- Vabi, M., Ngwasiri, C., Galega, P., Oyono, P.R., 2001. Devolution of forest management responsibilities to local communities in Cameroon. Context and implementation hurdles. WWF-CPO, Yaoundé.
- Vandergeest, P., 2003. Racialization and citizenship in Thai forest politics. *Society and Natural Resources* 16.
- Verhagen, H., Einthoven, C., 1993. Logging and conflicts in Cameroon. Friends of the Earth/Netherlands Committee of IUCN, Amsterdam.
- Wade, R., 1987. The management of common property resources. Finding a cooperative solution. *Research Observer* 2 (2), 190–210.
- Watts, M.J., 2000. Contested communities, malignant markets, and gilded governance: justice, resource extraction and environmental justice in the tropics. In: Zerner, C. (Ed.), *People, Plants, and Justice. The Politics of Nature Conservation*. Columbia University Press, New York, pp. 21–51.
- Wollenberg, E., Edmunds, D., Buck, L., 2000. Anticipating Change: Scenarios as tools for adaptive forest management. A guide. CIFOR, Bogor.
- Zerner, C., 2000. Toward a broader vision of justice and nature conservation. In: Zerner, C. (Ed.), *People, Plants, and Justice. The Politics of Nature Conservation*. Columbia University Press, New-York, pp. 3–20.
- Zinga, V., 1996. Halte à la braderie de nos forêts. *Le Messager* 223, 4–5.